

ABHAY RAMESH

Embedded Engineer

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PROFESSIONAL SUMMARY

Embedded Engineer with 1+ years of experience in embedded systems, firmware development, IoT solutions, networking and technical support across technology and IT service environments. Skilled in Embedded C, C++, Python, PIC/STM microcontrollers, Arduino, Raspberry Pi, communication protocols, debugging, troubleshooting, and system optimization. Experienced in developing biometric access, EV control, attendance, emergency communication, and hospital automation systems with strong expertise in hardware-software integration and MS SQL databases.

PROFESSIONAL EXPERIENCE

Embedded Engineer

Grapes Innovative Solutions | Infopark Koratty, Kerala, India

July 2025 – Present

- Achieved reliable embedded product deployment through firmware development, hardware-software integration, and system testing for client-specific solutions
- Developed embedded applications using microcontrollers, IoT components, and communication protocols to ensure stable device connectivity and real-time data transmission
- Improved system performance and debugging efficiency through firmware optimization, troubleshooting, and validation testing
- Collaborated with cross-functional teams to deliver scalable, production-ready embedded devices within project timelines
- Reduced testing downtime by streamlining debugging workflows and resolving integration issues efficiently

Customer Service Executive

Sutherland Global Services | Kakkanad, Kerala, India

August 2022 – March 2025

- Achieved consistent customer service performance by managing high-volume chat and email support while maintaining quality standards and response efficiency
- Resolved customer inquiries related to orders, accounts, and products through accurate communication and timely issue resolution
- Improved customer satisfaction by escalating complex issues to relevant teams and ensuring faster turnaround time
- Strengthened operational efficiency through effective ticket handling, issue tracking, and cross-functional coordination
- Developed strong communication, problem-solving, and customer relationship management skills in a fast-paced support environment

SKILLS

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|---------------------------------|-------------------------------|
| • Embedded Systems Development | • Raspberry Pi |
| • Firmware Development | • Communication Protocols |
| • Hardware-Software Integration | • System Integration |
| • IoT Solutions | • Debugging & Troubleshooting |
| • Embedded C | • Performance Optimization |
| • C++ | • RTSP Stream Processing |
| • Python | • MS SQL |
| • PIC Microcontrollers | • Database Management |
| • STM Microcontrollers | • Microsoft Excel |
| • Arduino | • Microsoft Office Suite |

PROJECTS

FaceID Access System

- Developed a deep-learning-based biometric security system using InsightFace and FAISS for real-time facial authentication
- Implemented RTSP stream processing and integrated MS SQL for access logging and audit management
- Designed a “Consensus Logic” engine to improve authentication reliability and system security

Electric Vehicle in College Campus

- Designed an EV-based campus transportation system focused on control systems, safety, and energy efficiency
- Worked on embedded control, component integration, and system performance evaluation

Attendance System

- Developed an automated attendance system using embedded hardware and software integration
- Improved attendance tracking efficiency through automated data processing and monitoring

Emergency Call System

- Built a real-time emergency alert and communication system for rapid response applications
- Improved emergency notification reliability through embedded system integration

Hospital Automation with Database

- Developed a hospital automation system integrating embedded devices with database management
- Streamlined operational workflows through automation and centralized data handling

EDUCATION

Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering

Ponnaiyah Ramajayam Institute of Science & Technology | 2018 – 2022

CERTIFICATION

Embedded and IoT Engineer

IPCS Kochi | March 2025 – July 2025

- Completed intensive training in Embedded Systems and IoT technologies
- Gained practical exposure in microcontrollers, automation systems, and embedded development

OTHER EXPERIENCE

Freelance Electrician

Self-Employed | Kerala, India | Mar 2025 – July 2025

- Performed electrical installation, wiring, maintenance, and troubleshooting for residential and commercial projects.
- Ensured safe and timely completion of electrical repair and maintenance work based on client requirements.

LANGUAGES

English | Malayalam | Hindi